MATH0110 — Hard Problem Set

Analysis for Joint Honours · UCL

> **10 problems** spanning every major topic. Full solutions included.
> Topics: Supremum · Sequences · Series · Power Series · - Limits · Continuity · Differentiation · MVT · Integration · EVT

Problem 1 — Supremum

Let $A, B \subseteq \mathbb{R}$ be non-empty bounded sets. Define

$$A + B = \{a + b : a \in A,\ b \in B\}.$$

Prove that $\sup(A + B) = \sup A + \sup B$.

Solution

Let $S = \sup A$ and $T = \sup B$. We verify both conditions of the supremum definition for $S + T$.

$S + T$ is an upper bound. For any $a + b \in A + B$, we have $a \le S$ and $b \le T$, so:

$$a + b \le S + T.$$

$S + T$ is the least upper bound. Let $\varepsilon > 0$. By the definition of supremum applied to A, there exists $a \in A$ with $a > S - \varepsilon/2$. Similarly, there exists $b \in B$ with $b > T - \varepsilon/2$. Then $a + b \in A + B$ and:

$$a + b > S + T - \varepsilon.$$

So $S + T - \varepsilon$ is not an upper bound of $A + B$. Since $\varepsilon > 0$ was arbitrary, no number strictly less than $S + T$ can be an upper bound, so $S + T = \sup(A + B)$. ■

Problem 2 — Sequences (Convergence from Definition)

Let (a_n) be a sequence with $a_n \ge 0$ for all n, and suppose $a_n \to L$. Using only the definition of convergence, prove that $\sqrt{a_n} \to \sqrt{L}$.

Treat the cases $L = 0$ and $L > 0$ separately.

Solution

Case 1: $L = 0$.

Let $\varepsilon > 0$. Since $a_n \to 0$, there exists $N \in \mathbb{N}$ such that for all $n \ge N$:

$$|a_n| < \varepsilon^2.$$

Then $|\sqrt{a_n} - 0| = \sqrt{a_n} < \varepsilon$. Done.

Case 2: $L > 0$.

Use the rationalisation identity:

$$|\sqrt{a_n} - \sqrt{L}| = \frac{|a_n - L|}{\sqrt{a_n} + \sqrt{L}} \le \frac{|a_n - L|}{\sqrt{L}}.$$

Let $\varepsilon > 0$. Since $a_n \to L$, there exists $N \in \mathbb{N}$ such that for all $n \ge N$:

$$|a_n - L| < \varepsilon\sqrt{L}.$$

Then:

$$|\sqrt{a_n} - \sqrt{L}| \le \frac{|a_n - L|}{\sqrt{L}} < \frac{\varepsilon\sqrt{L}}{\sqrt{L}} = \varepsilon. \quad \blacksquare$$

Problem 3 — Series (True or False)

Determine whether each statement is true or false. Prove it or give a counterexample.

(i) If $\sum a_n$ converges absolutely, then $\sum \dfrac{a_n}{1 + a_n^2}$ converges.

(ii) If $\sum a_n$ and $\sum b_n$ are both conditionally convergent, then $\sum(a_n + b_n)$ is conditionally convergent.

Solution

(i) TRUE.

Since $\sum |a_n|$ converges, we have $a_n \to 0$, so there exists N such that $|a_n| < 1$ for all $n \ge N$. For such n, since $1 + a_n^2 \ge 1$:

$$\left| \frac{a_n}{1 + a_n^2} \right| \le |a_n|.$$

By the Basic Comparison Test applied to $\sum |a_n|$, the series $\sum \frac{a_n}{1+a_n^2}$ converges absolutely, hence converges. $\blacksquare$

(ii) FALSE.

Let $a_n = \dfrac{(-1)^n}{\sqrt{n}}$ and $b_n = \dfrac{(-1)^{n+1}}{\sqrt{n}} = -a_n$.

- Both $\sum a_n$ and $\sum b_n$ converge conditionally by the Alternating Series Test (since $1/\sqrt{n}$ decreases to 0), but $\sum 1/\sqrt{n}$ diverges (p-series, $p = 1/2 \le 1$).
- However, $a_n + b_n = 0$ for all n, so $\sum(a_n + b_n) = 0$, which converges **absolutely**.

This is not conditionally convergent (a series is conditionally convergent iff it converges but does not converge absolutely), so the statement is false. ■

Problem 4 — Power Series (Radius & Endpoints)

Find all $x \in \mathbb{R}$ for which the series

$$\sum_{n=0}^{\infty} \frac{3^n}{(n+2)^2}\, x^n$$

converges.

Solution

Let $a_n = \dfrac{3^n}{(n+2)^2} x^n$. Apply the Ratio Test:

$$\left|\frac{a_{n+1}}{a_n}\right| = |3x| \cdot \frac{(n+2)^2}{(n+3)^2} \longrightarrow 3|x| \quad \text{as } n \to \infty.$$

By the Ratio Test, the series converges if $3|x| < 1$ (i.e. $|x| < 1/3$) and diverges if $3|x| > 1$ (i.e. $|x| > 1/3$). It remains to check the endpoints.

Endpoint $x = 1/3$:

$$\sum_{n=0}^{\infty} \frac{3^n}{(n+2)^2} \cdot \frac{1}{3^n} = \sum_{n=0}^{\infty} \frac{1}{(n+2)^2},$$

which converges (p-series, $p = 2 > 1$).

Endpoint $x = -1/3$:

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+2)^2},$$

which converges absolutely by the same p-series argument.

Interval of convergence: $\boxed{\left[-\tfrac{1}{3},\ \tfrac{1}{3}\right]}$. ■

Problem 5 — ε-δ Limit

Prove using the ε-δ definition that

$$\lim_{x \to 2} (x^2 - 5x + 3) = -3.$$

Solution

We need to show $|(x^2 - 5x + 3) - (-3)| < \varepsilon$, i.e.:

$$|x^2 - 5x + 6| = |(x-2)(x-3)| < \varepsilon.$$

Finding δ. Restrict $|x-2| < 1$, so $x \in (1,3)$. Then $|x-3| < 2$, and so:

$$|(x-2)(x-3)| \le 2|x-2|.$$

This suggests taking $\delta = \min\left\{1, \ \dfrac{\varepsilon}{2}\right\}$.

Proof. Let $\varepsilon > 0$ and set $\delta = \min\{1, \varepsilon/2\}$. Suppose $0 < |x-2| < \delta$. Then:

$$|(x^2 - 5x + 3) - (-3)| = |(x-2)(x-3)| \le 2|x-2| < 2 \cdot \frac{\varepsilon}{2} = \varepsilon. \quad \blacksquare$$

Problem 6 — Continuity (Hölder Condition)

A function $f : \mathbb{R} \to \mathbb{R}$ satisfies

$$|f(x) - f(y)| \le |x-y|^{1/2} \quad \text{for all } x, y \in \mathbb{R}.$$

Prove that f is uniformly continuous on $\mathbb{R}$, and deduce it is continuous.

Solution

Uniform continuity. Let $\varepsilon > 0$. Set $\delta = \varepsilon^2$. For *any* $x, y \in \mathbb{R}$ with $|x-y| < \delta$:

$$|f(x) - f(y)| \le |x-y|^{1/2} < \delta^{1/2} = \varepsilon.$$

Note that δ depends only on ε, not on the specific choice of x or y. This is precisely the definition of uniform continuity. $\blacksquare$

Continuity. Fix any $a \in \mathbb{R}$. For any $\varepsilon > 0$, the same $\delta = \varepsilon^2$ works for the pointwise continuity condition at a (it is a special case of uniform continuity). Hence f is continuous at every $a \in \mathbb{R}$. $\blacksquare$

Problem 7 — Derivative from Definition

Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable at a, with $f(a) > 0$. Define $g(x) = \sqrt{f(x)}$. Prove directly from the definition of derivative — without using the chain rule — that g is differentiable at a and

$$g'(a) = \frac{f'(a)}{2\sqrt{f(a)}}.$$

Solution

We compute the difference quotient for g:

$$\frac{g(a+h)-g(a)}{h} = \frac{\sqrt{f(a+h)}-\sqrt{f(a)}}{h}.$$

Rationalise by multiplying numerator and denominator by $\sqrt{f(a+h)}+\sqrt{f(a)}$:

$$= \frac{f(a+h)-f(a)}{h} \cdot \frac{1}{\sqrt{f(a+h)}+\sqrt{f(a)}}.$$

Now take the limit as $h \to 0$:

- Since f is differentiable at a, it is continuous there, so $f(a+h) \to f(a)$ as $h \to 0$.
- Therefore $\sqrt{f(a+h)} \to \sqrt{f(a)}$, and since $f(a) > 0$:

$$\frac{1}{\sqrt{f(a+h)}+\sqrt{f(a)}} \longrightarrow \frac{1}{2\sqrt{f(a)}}.$$

- Also $\dfrac{f(a+h)-f(a)}{h} \to f'(a)$.

By the limit laws:

$$g'(a) = \lim_{h\to 0} \frac{g(a+h)-g(a)}{h} = f'(a) \cdot \frac{1}{2\sqrt{f(a)}} = \frac{f'(a)}{2\sqrt{f(a)}}. \quad \blacksquare$$

Problem 8 — Mean Value Theorem Application

Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b). Suppose $f'(x) \neq 0$ for all $x \in (a,b)$. Prove that f is injective on $[a,b]$.

Solution

We prove the contrapositive: if f is not injective, then there exists $x \in (a,b)$ with $f'(x) = 0$.

Suppose there exist distinct $\alpha, \beta \in [a,b]$ with $f(\alpha) = f(\beta)$. Without loss of generality, $\alpha < \beta$. Since f is continuous on $[a,b]$ and differentiable on (a,b), it is continuous on $[\alpha,\beta]$ and differentiable on (α,β). Moreover $f(\alpha) = f(\beta)$.

By **Rolle's Theorem**, there exists $c \in (\alpha,\beta) \subseteq (a,b)$ such that:

$$f'(c) = 0.$$

This contradicts the assumption $f'(x) \neq 0$ for all $x \in (a,b)$. Therefore f must be injective. $\blacksquare$

Problem 9 — Riemann Integration

Let $f : [a,b] \to \mathbb{R}$ be bounded and integrable with integral I. Prove that for every $\varepsilon > 0$ there exists a partition P of $[a,b]$ such that

$$U(f,P) - L(f,P) < \varepsilon.$$

Solution

Fix $\varepsilon > 0$.

Finding P_1. Since $I = \overline{\int_a^b} f = \inf_P U(f,P)$, and $I + \varepsilon/2 > I$, the number $I + \varepsilon/2$ is not a lower bound for the upper sums. Hence there exists a partition P_1 such that:

$$U(f,P_1) < I + \frac{\varepsilon}{2}.$$

Finding P_2. Since $I = \underline{\int_a^b} f = \sup_P L(f,P)$, and $I - \varepsilon/2 < I$, the number $I - \varepsilon/2$ is not an upper bound for the lower sums. Hence there exists a partition P_2 such that:

$$L(f,P_2) > I - \frac{\varepsilon}{2}.$$

Taking the common refinement. Let $P = P_1 \cup P_2$. Adding points to a partition can only decrease upper sums and increase lower sums, so:

$$U(f,P) \le U(f,P_1) < I + \frac{\varepsilon}{2}, \qquad L(f,P) \ge L(f,P_2) > I - \frac{\varepsilon}{2}.$$

Therefore:

$$U(f,P) - L(f,P) < \left(I + \frac{\varepsilon}{2}\right) - \left(I - \frac{\varepsilon}{2}\right) = \varepsilon. \quad \blacksquare$$

Problem 10 — EVT + Derivative (Theorem Proof)

Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable everywhere. Let $a < b$ with $f'(a) > 0 > f'(b)$. Prove that there exists $c \in (a,b)$ such that $f'(c) = 0$.

Note: You may NOT simply apply IVT to f' — explain why that would be wrong.

Solution

Why IVT fails here. The IVT requires the function to be *continuous* on the interval. We only know f is differentiable, which does not guarantee f' is continuous. So applying IVT directly to f' is not justified.

Step 1: f exceeds $f(a)$ somewhere inside (a,b).

Since $f'(a) > 0$, by the definition of derivative:

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = f'(a) > 0.$$

Take $\varepsilon_1 = f'(a)/2 > 0$. There exists $\delta_1 > 0$ (small enough that $a + \delta_1 < b$) such that for $0 < h < \delta_1$:

$$\frac{f(a+h) - f(a)}{h} > \frac{f'(a)}{2} > 0.$$

In particular, taking $h = \delta_1/2$, we get $d_1 = a + \delta_1/2 \in (a, b)$ with $f(d_1) > f(a)$.

Step 2: f exceeds $f(b)$ somewhere inside (a, b).

Since $f'(b) < 0$, similarly there exists $\delta_2 > 0$ (small enough that $b - \delta_2 > a$) such that for $0 < h < \delta_2$:

$$\frac{f(b-h) - f(b)}{-h} = \frac{f(b + (-h)) - f(b)}{-h} < 0 \implies f(b-h) > f(b).$$

Setting $d_2 = b - \delta_2/2 \in (a, b)$ gives $f(d_2) > f(b)$.

Step 3: Apply the Extreme Value Theorem.

Since f is differentiable on $\mathbb{R}$, it is continuous on $[a, b]$. By the **Extreme Value Theorem**, f attains its maximum on $[a, b]$ at some point $c \in [a, b]$.

From Steps 1 and 2, $f(d_1) > f(a)$ and $f(d_2) > f(b)$, so the maximum is not attained at either endpoint. Therefore $c \in (a, b)$.

Step 4: Conclude $f'(c) = 0$.

Since $c \in (a, b)$ is an interior point and $f(c) \geq f(x)$ for all $x \in [a, b]$, we show $f'(c) = 0$ from the definition:

- For $h > 0$ small: $f(c+h) \leq f(c)$, so $\frac{f(c+h) - f(c)}{h} \leq 0$. Taking $h \to 0^+$ gives $f'(c) \leq 0$.
- For $h < 0$ small: $f(c+h) \leq f(c)$, so $\frac{f(c+h) - f(c)}{h} \geq 0$. Taking $h \to 0^-$ gives $f'(c) \geq 0$.

Since $f'(c)$ exists and satisfies both $f'(c) \leq 0$ and $f'(c) \geq 0$, we conclude $f'(c) = 0$. $\blacksquare$

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